

The empirical recurrence for OEIS A251226, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A251226 records “Empirical recurrence of order 54” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 128$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 56$. Its last coefficient is $c_{54} = 2232000000 \neq 0$, so no further tail satisfies anything shorter; any order-54 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251226 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with no 2×2 subblock having the minimum of its diagonal elements greater than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

7275, 453273, 27662994, 1697323707, 103986727042, 6373427948731, 390586327228567, 23937316421928928, ...

The entry states:

Empirical recurrence of order 54 (see link above).

As of the “Last modified” line on the live entry (Mar 15 2023 11:13:50, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 128$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 128$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 54: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 54.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 256$ exact terms from $n = 2$ on, it returns order 54 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	34	c_{15}	−1915847851199217	c_{29}	8386121557483387550	c_{43}	−39833960901036025
c_2	1980	c_{16}	−2745968113317754	c_{30}	−53595682351374577296	c_{44}	−7736217862148976
c_3	−6208	c_{17}	24384967466752051	c_{31}	−6325037005833278777	c_{45}	39457671212398594
c_4	−793193	c_{18}	−2104693897112216	c_{32}	62675306522495696852	c_{46}	3288288518260903
c_5	−1178400	c_{19}	−183244863265275870	c_{33}	6413536450434696407	c_{47}	−2348847172137039
c_6	136845589	c_{20}	167313643325131020	c_{34}	−56571493501597109540	c_{48}	−286834426138181
c_7	282954809	c_{21}	877727041911766836	c_{35}	−10520065121147427810	c_{49}	82069690748093
c_8	−13184042821	c_{22}	−1334560766297236963	c_{36}	37084679815446162107	c_{50}	15908948100889
c_9	−12481473905	c_{23}	−2778154245528491980	c_{37}	12921993877442780655	c_{51}	−582327299180
c_{10}	781426508243	c_{24}	5882802252855904661	c_{38}	−14591541166245292895	c_{52}	−407464559180
c_{11}	−695430590958	c_{25}	5910513295499464229	c_{39}	−7718884754284429966	c_{53}	−21448558800
c_{12}	−26694422741339	c_{26}	−17070681515414188634	c_{40}	2985907858989019981	c_{54}	2232000000
c_{13}	74187239806833	c_{27}	−8516999716367398394	c_{41}	2350992584853791051		
c_{14}	416415264256190	c_{28}	35110542930181023170	c_{42}	−253491761416963288		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 56$, and it is the recurrence OEIS A251226 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{54} - c_1 x^{54-1} - \dots - c_{54}$. Its constant term is $-c_{54} = -2232000000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 54 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 54, so $\deg q = 54$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 54, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 54, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 2232000000.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-54 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251226.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.